Algebra 2 Review: Linear Functions & Systems

Answer Key

Station 1: Equations of Lines, Slope, Parallel and Perpendicular

1. $y = -2x - 1$
$m = \dfrac{-9-5}{4-(-3)} = -2$, *then* $5 = -2(-3) + b$

2. $f(x) = 2x + 3$; $f(20) = 43$
constant rate: $\dfrac{13-7}{5-2} = 2$

3. $y + 2 = -\dfrac{3}{4}(x - 8)$; $y = -\dfrac{3}{4}x + 4$
point-slope uses $y - y_1 = m(x - x_1)$, *so* $-(-2)$ *becomes* $+2$

4. $5x - 2y = 16$
$y + 3 = \dfrac{5}{2}(x - 2) \Rightarrow 2y + 6 = 5x - 10$

5. (a) $y = -\dfrac{2}{5}x + 1$; (b) $y = \dfrac{5}{2}x - \dfrac{27}{2}$
slope of L is $-\dfrac{2}{5}$*; perpendicular slope is the negative reciprocal,* $\dfrac{5}{2}$

6. $y = -\dfrac{3}{2}x + 3$; x-intercept 2
rise -3 *for run* 2*;* y*-intercept* 3

7. ℓ_1, ℓ_2 parallel (both slope 2); ℓ_1, ℓ_3 perpendicular; ℓ_2, ℓ_3 perpendicular
slopes are 2, 2, *and* $-\dfrac{1}{2}$

8. $C(m) = 28m + 45$; $C(18) = \$549$; $m = 20$ months
$28m + 45 = 605$

9. $k = -1$
need slope -3*:* $\dfrac{-5-4}{2-k} = -3$

10. $C(g) = 30$ for $g \le 5$; $C(g) = 30 + 8(g - 5)$ for $g > 5$; $C(9) = \$62$
4 GB over the limit: $30 + 8(4)$

Station 2: Linear Models, Scatter Plots, and Best-Fit Lines

1. $P(t) = 150t + 8400$: grows 150 people per year; $P(18) = 11{,}100$
$\dfrac{9600 - 8400}{8} = 150$

2. Slope -35: drains 35 gal per minute; intercept 560: starts with 560 gal; empty at $t = 16$ min
$560 - 35t = 0$

3. $\hat{y} = 4.1x + 9.9$; $r \approx 0.98$: strong positive correlation
round only at the end

4. Strong negative correlation; slope -2: loses about \$2,000 per year; intercept 20: about \$20,000 new; $y(7) = 6$, about \$6,000
7 years is inside the data range

5. predicted 68.7; residual $69 - 68.7 = 0.3$: the actual value is 0.3 above the line
residual = actual − predicted

6. Strong negative linear relationship; $r^2 = 0.8464$, so about 85% of the variation in sleep is explained by the line; correlation does not prove causation

7. $E(t) = 1000 + 300t$ for $0 \le t \le 2$; $E(t) = 2000 - 200t$ for $2 < t \le 8$; $t = 4$ hours
peak is 1,600 ft at $t = 2$

8. A grows 2.1 cm per day, B grows 1.8; A is faster by 0.3 cm per day
$\dfrac{12.4 - 4}{6 - 2} = 2.1$

9. $F = \dfrac{9}{5}C + 32$; they agree at $-40°$
solve $\dfrac{9}{5}C + 32 = C$

10. $R(250) = -0.3$ s: impossible; the model should not be extended far beyond the data

Station 3: Systems of Equations and Inequalities

1. $y = -x + 5$ and $y = 2x - 1$; solution $(2, 3)$
   $3 = -2 + 5$ *and* $3 = 2(2) - 1$
2. $(3, 2)$
   $3x + 4(2x - 4) = 17 \Rightarrow 11x = 33$
3. $\left(3, -\frac{5}{2}\right)$
   double the first equation: $6x + 4y = 8$
4. (a) no solution (parallel lines, different intercepts); (b) infinitely many (same line)
   multiply (a)'s second equation by -2*:* $-2x + 6y = -8 \neq 9$
5. boat 10 mph, current 2 mph
   $b - c = 8$ *and* $b + c = 12$
6. $n = 48$ items, where cost and revenue both equal \$1,920; selling more earns a profit
7. $y < \frac{1}{2}x + 2$ and $y \geq -x + 1$; $(2, 4)$: $4 < 3$ is false, so it is not a solution
   dashed means strict; test a point on each side
8. only $(3, 1)$; $(0, 0)$ satisfies $x + 2y < 6$ but $3(0) - 0 \geq 2$ is false
   a point must satisfy every inequality
9. $2c + 3b \geq 60$, $c + b \leq 30$, $c \geq 0$, $b \geq 0$; $(15, 10)$ works (\$60, 25 items); $(10, 10)$ fails (\$50 < \$60)
10. $k = 3$; with $k = 3$ the second equation must read $6x + 9y = 15$
    same slope: $\frac{2}{6} = \frac{k}{9}$*;* $3 \cdot 5 = 15$

Station 4: Linear Programming and Three-Variable Systems

1. $(0, 0)$, $(8, 0)$, $(6, 2)$, $(0, 4)$
   solve $x + y = 8$ *and* $x + 3y = 12$ *together*
2. maximum 28 at $(6, 2)$
   values at the vertices: 0, 24, 28, 20
3. minimum 18 at $(0, 6)$
   vertices $(0, 6)$*,* $(4, 2)$*,* $(8, 0)$ *give* 18, 22, 32
4. $2c + p \leq 40$, $c + 3p \leq 45$, $c, p \geq 0$; 15 cakes and 10 pies give \$270
   vertices $(0, 0), (20, 0), (0, 15), (15, 10)$ *give \$0, \$240, \$135, \$270*
5. maximum 12 at $(4, 4)$; minimum 3 at $(1, 1)$
   check all four vertices: 3, 11, 12, 8
6. $(3, 1, -2)$
   add eqs. 1 and 3 to get $2x + 3y = 9$*; eliminate z again from eqs. 1 and 2:* $5x + 2y = 17$
7. 10 nickels, 15 dimes, 35 quarters
   in cents: $5n + 10d + 25q = 1075$
8. (a) no solution (the first two planes are parallel); (b) infinitely many: $(x,\ x - 1,\ 5 - 2x)$
   twice eq. 1 is 8*, not* 9 *in (a); in (b) the equations agree*
9. $y = x^2 - 2x + 3$
   $a + b + c = 2$*,* $4a + 2b + c = 3$*,* $9a + 3b + c = 6$*; subtract to get* $3a + b = 1$ *and* $5a + b = 3$
10. maximum 12 at $(6, 2)$ and $(0, 4)$ and at every point between them, such as $(3, 3)$ (infinitely many)
    the objective line is parallel to the edge $x + 3y = 12$